

METHODOLOGY — ML / DL & MODELLING RESEARCH

Arkam B.H.M. (214019K) | Integrity-first modelling pipeline for Big Onion yield under severe label scarcity (n = 28 district-years)

1 DATA FOUNDATION

Reduce four independent streams to one common district-season-year key

DCS Yield Panel

87 genuine month-records
Yield = production / extent
28 district-years, 2019-25

NASA POWER

37,988 daily records
4 district centroids
2000-2025 climatology

SoilGrids 250 m

pH, organic carbon
clay %, sand %
static per district

MODIS / GEE

NDVI, EVI, NDWI, LST
server-side seasonal
reduction (scope-limited)

2 INTEGRITY GATE

Nothing is modelled until every target value traces to a real DCS record

Provenance Audit

50 / 124 rows flagged
source = synthetic
target rebuilt from source

Plausibility Filter

6 impossible records cut
(worst 442 MT/ha)
thousands-separator parse fix

Leakage Removal

4 look-ahead leaks
1981-2018 climatology base
lagged-yield terms dropped

Extent Weighting

Cells span 3.5-1,765 ha
weight = sqrt(area), mean 1
weighted + unweighted reported

3 ATTAINABILITY

Establish what the panel can support before any model is ranked

Variance Decomposition

Between years 63.6 %
Between districts 2.2 %
Residual 34.2 %

Measurement-Error Model

Weighted spread of the month-records per cell
53.6 % of within-year var

Attainable Ceiling

Implied LOYO R2 = 0.162
Proposal target 0.75 was unreachable by any model

4 MODEL FAMILY

Nine predictors trained under one identical protocol and seed

Classical

Random Forest, XGBoost
SVR (RBF), depth + L1/L2
capped, nested tuning

Deep Sequence

LSTM, BiLSTM, 1D-CNN,
hybrid CNN-LSTM (44,929 p)
weights reset every fold

Symbolic + Stack

gplearn GP over top-5
SHAP predictors; convex
simplex-constrained blend

PADR (contribution)

Phenology-Aligned Differentiable
Response - 17 parameters,
agronomic constants ESTIMATED

5 HONEST EVALUATION

Every fitted quantity re-estimated inside the fold; nothing pooled

Leave-One-Year-Out

7 folds x 4 held-out cells
Nested inner tuning
Wilcoxon paired residuals

Mechanism Ablations

7 arms isolate each claim
Learned vs fixed physics:
dR2 = +0.566

Power + Identifiability

Detects weather signal at
6 % of variance; 5 / 7
constants recovered

Blocked Conformal

Year-blocked cross-conformal
Coverage 0.911 vs 0.90
Half-width +/- 15.3 MT/ha